

01 Vectors

1.8 Vector Norm

Mapping vectors of every direction onto a single comparable number



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

What You Will Learn

1

Understand the core idea of a vector norm: a unified rule that maps vectors of any direction onto a single comparable scalar

3

Master the general L_p norm: a continuous generalization spanning L_1 to L_∞ , with $p \geq 1$

5

Understand how vector norms relate to distance metrics — Euclidean distance, Manhattan distance, Chebyshev distance

7

See the special case $0 < p < 1$: the contour turns star-shaped, and it no longer satisfies the definition of a norm

2

Master three basic norms: L_1 (sum of absolute values), L_2 (Pythagorean theorem, Euclidean distance), L_∞ (largest absolute component)

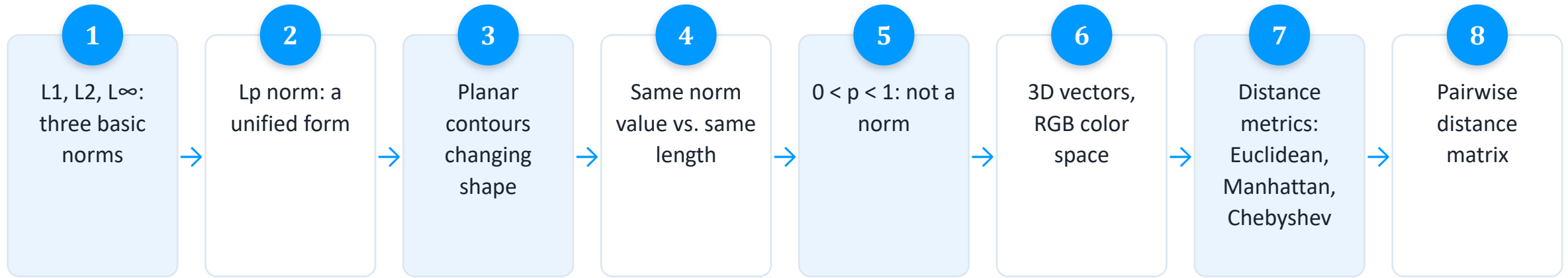
4

Build geometric intuition through planar iso-distance lines and spatial iso-distance surfaces for different norms

6

Understand the structure of a pairwise distance matrix: storing distances between every pair of points in one matrix

How the Ideas Connect



How Do We Compare the "Size" of Vectors?

$|x|$ = distance from x to the origin; $\|\mathbf{x}\|$ = the "distance" from a high-dimensional vector to the origin

- On a number line, 5 is larger than 4; even with different signs, say -5 and 4 , we can compare after taking absolute values
- The absolute value is really just the distance between a number and the origin — an idea that extends naturally to 2D, 3D, and beyond
- To make high-dimensional vectors comparable, we introduce the vector norm: a rule that first maps a vector to a scalar, then compares

L1 Norm: Sum of the Absolute Values of the Components

$$\|\mathbf{x}\|_1 = |x_1| + |x_2| + \dots + |x_n|$$

- The L1 norm is also called the Manhattan distance or city block distance
- For a 2D vector $\mathbf{x} = [x_1, x_2]^T$, the L1 norm is $\|\mathbf{x}\|_1 = |x_1| + |x_2|$
- Example: $\mathbf{a} = [3, -4]^T$ has L1 norm $|3| + |-4| = 7$; $\mathbf{b} = [0, 5]^T$ has L1 norm $|0| + |5| = 5$, so by the L1 norm, $\mathbf{a}$ is larger

Computing the L1 Norm with NumPy

Two equivalent ways to reach the same result

- Define vectors $a = [3, -4]$ and $b = [0, 5]$ with `np.array()`
- `np.linalg.norm(a, ord=1)` computes the L1 norm directly; `ord=1` tells NumPy to sum absolute values
- `np.sum(np.abs(a))` is the equivalent manual route: take the absolute value of each component, then sum — again 7
- The same steps applied to b give 5, confirming that a is larger under the L1 norm

```
## 初始化
import numpy as np

## 定义向量
a = np.array([3, -4])
b = np.array([0, 5])

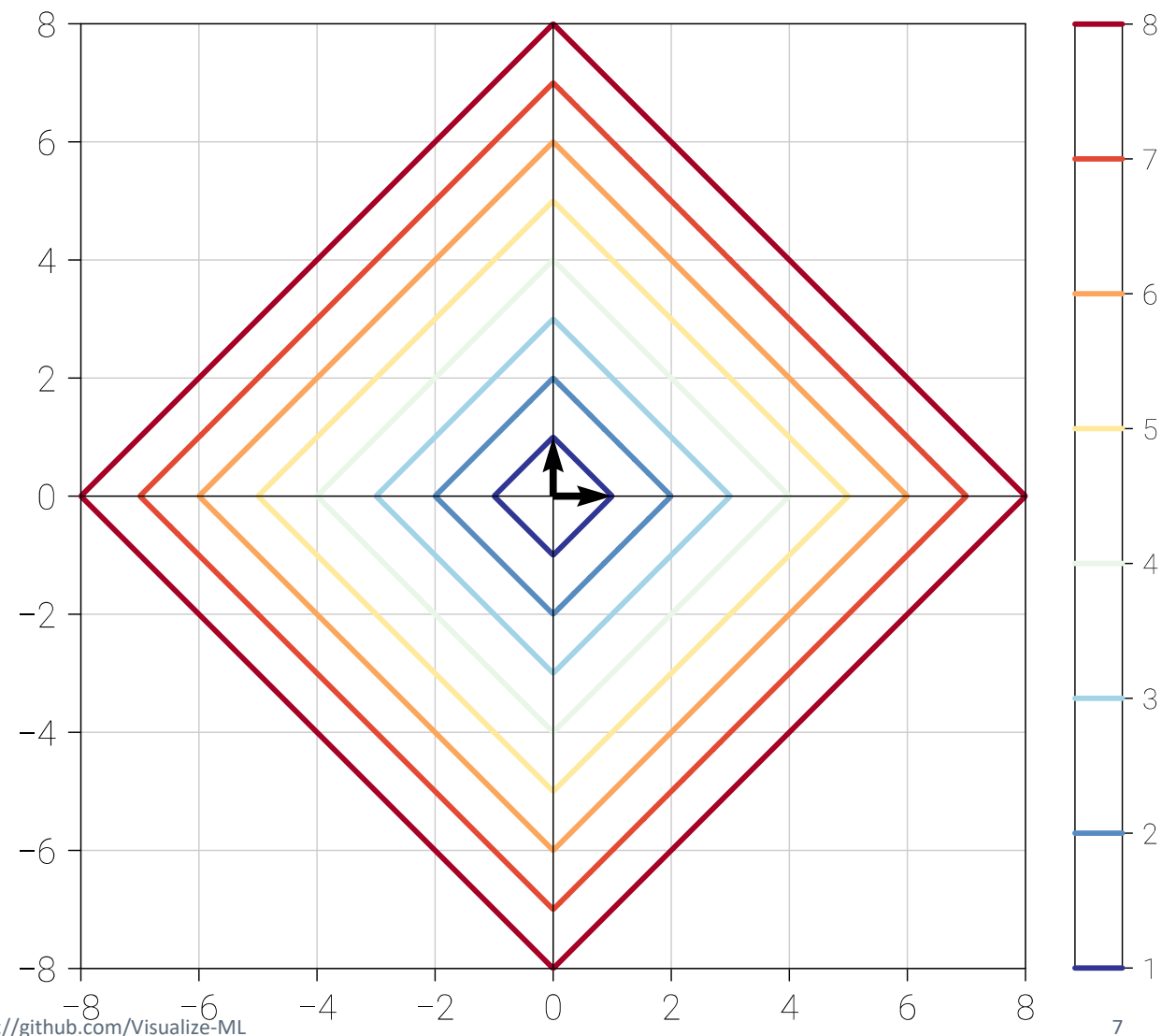
## 计算 L1 范数
a L1_a = np.linalg.norm(a, ord=1)
b np.sum(np.abs(a))

L1_b = np.linalg.norm(b, ord=1)
np.sum(np.abs(b))
```



Figure 1 · Planar Iso-Distance Line of the 2D L1 Norm

- The L1 norm's iso-distance line is a rotated square (a special rhombus), satisfying $|x_1| + |x_2| = c$, $c \geq 0$
- The black arrows are $\mathbf{e}_1 = [1, 0]^T$ and $\mathbf{e}_2 = [0, 1]^T$, each with an L1 norm of 1
- In the first quadrant ($x_1 > 0$, $x_2 > 0$), vectors with L1 norm 1 satisfy $x_1 + x_2 = 1$ — try writing the expressions for the other quadrants



Plotting the L1 Norm's Planar Iso-Distance Line

*meshgrid lays down the grid, contour
draws the lines*

- `np.meshgrid()` generates a grid of coordinates covering the plane, ready for a norm computation at every point
- `zz = np.abs(xx) + np.abs(yy)` computes the L1 norm at each grid point
- `plt.quiver()` draws basis vectors `e1`, `e2`; `angles='xy'` and `scale_units='xy'` keep arrow length tied to the axis units
- `plt.contour(xx, yy, zz, levels=np.arange(1, 9))` draws L1-norm contours from 1 to 8, with `cmap` controlling the color gradient

```
## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 生成数据
xx, yy = np.meshgrid(np.linspace(-8, 8, 1000),
                     np.linspace(-8, 8, 1000))

## 计算 L1 范数
zz = np.abs(xx) + np.abs(yy) # L1 范数计算公式

## 定义标准基向量
e1 = np.array([1, 0])
e2 = np.array([0, 1])

## 可视化
plt.figure(figsize=(6, 6))

# 绘制标准基向量 e1 和 e2
plt.quiver(0, 0, e1[0], e1[1],
           angles='xy', scale_units='xy',
           scale=1, color=[0, 0, 0],
           label='e1', zorder=1000)

plt.quiver(0, 0, e2[0], e2[1],
           angles='xy', scale_units='xy',
           scale=1, color=[0, 0, 0],
           label='e2', zorder=1000)

# 绘制 L1 范数的等距线
contour = plt.contour(xx, yy, zz,
                     levels=np.arange(1, 9),
                     cmap='RdYlBu_r')

# 添加 colorbar
cbar = plt.colorbar(contour)

# 设置坐标轴范围
plt.xlim(-8, 8); plt.ylim(-8, 8)

# 绘制网格和坐标轴
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.xticks(np.arange(-8, 9, 2)); plt.yticks(np.arange(-8, 9, 2))
plt.grid(True, which='both', linestyle='--',
        linewidth=0.5, color='0.8')

# 设置等比例显示
plt.gca().set_aspect('equal', adjustable='box')
```

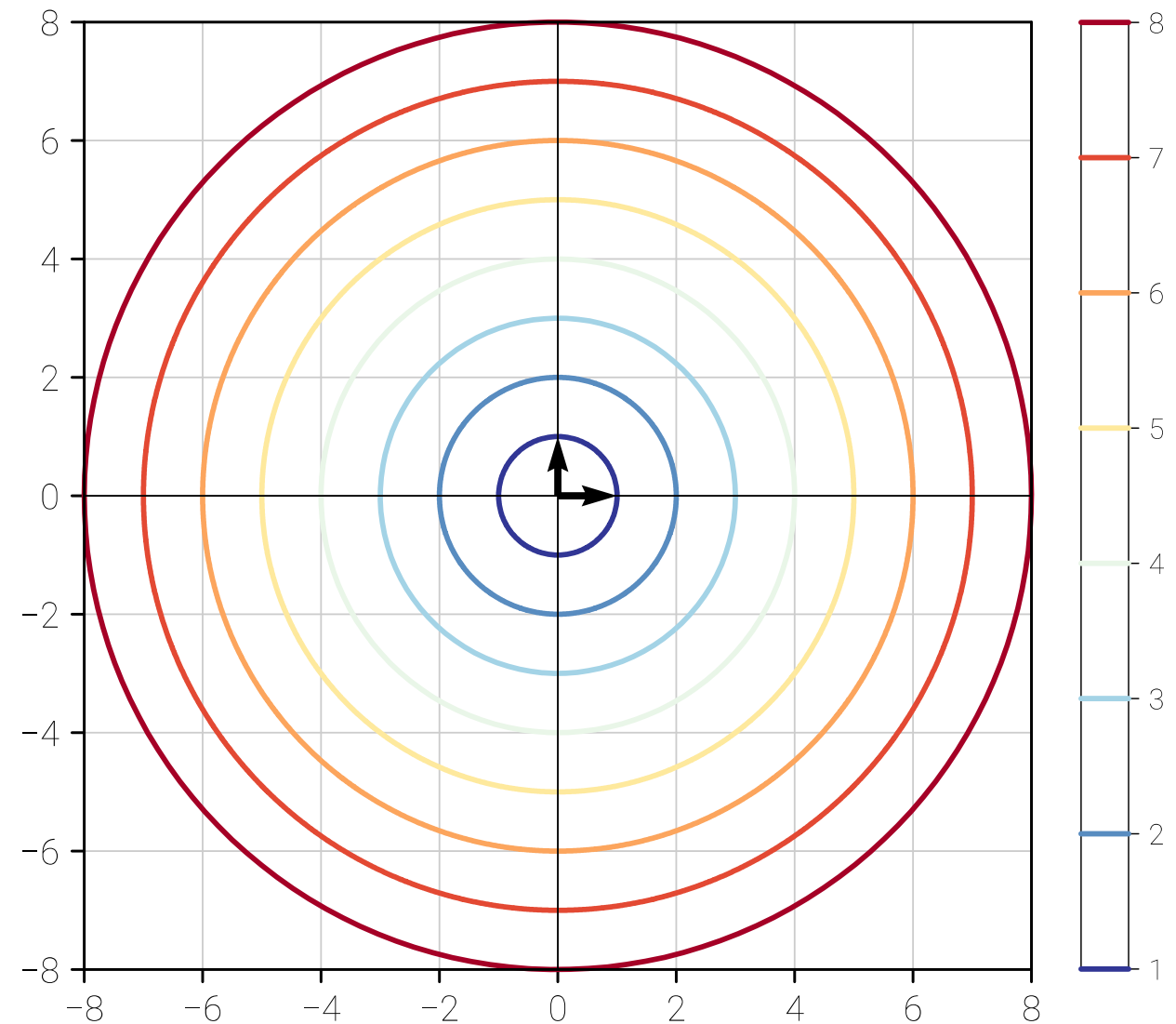
L2 Norm: the Euclidean Norm, a Vector's "Length"

$$\|\mathbf{x}\|_2 = \sqrt{(x_1^2 + x_2^2 + \dots + x_n^2)}$$

- The L2 norm is also called the Euclidean norm, or magnitude — the most familiar measure of length and distance in the physical world
- For a 2D vector $\mathbf{x} = [x_1, x_2]^T$, the L2 norm is $\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2}$
- Example: $\mathbf{a} = [3, -4]^T$ has L2 norm $\sqrt{3^2 + (-4)^2} = 5$; $\mathbf{b} = [0, 5]^T$ has L2 norm $\sqrt{0^2 + 5^2} = 5$ — the two are equal

Figure 2 · Planar Iso-Distance Line of the 2D L2 Norm

- The L2 norm's iso-distance line is a perfect circle, satisfying $x_1^2 + x_2^2 = c^2$, $c \geq 0$
- LA_01_08_03.ipynb computes and compares the L2 norms of $\mathbf{a}$, $\mathbf{b}$; LA_01_08_04.ipynb plots this figure — both are short and worth studying on your own



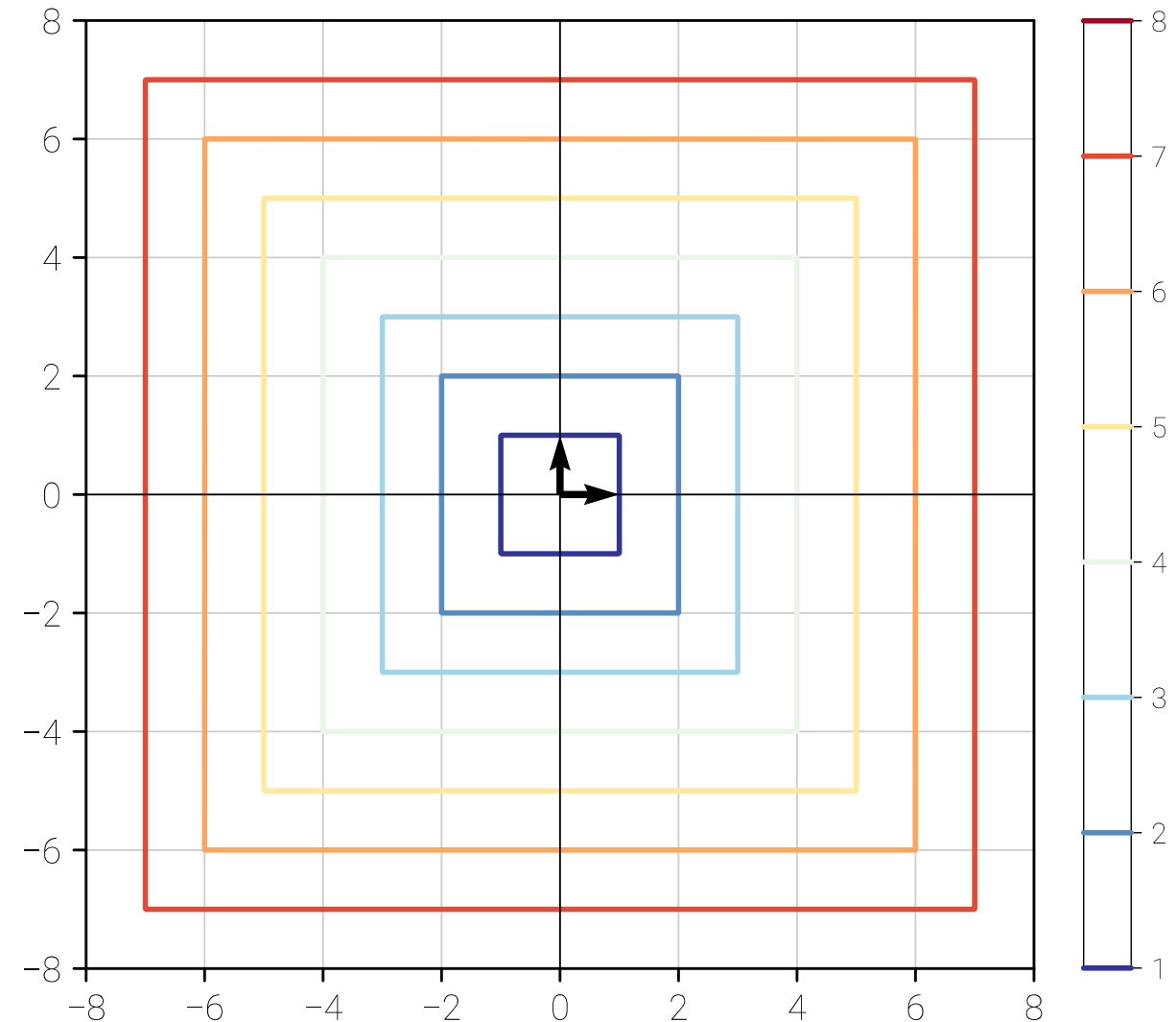
L^∞ Norm: the Largest Absolute Component

$$\|\mathbf{x}\|_\infty = \max(|x_1|, |x_2|, \dots, |x_n|)$$

- The L^∞ norm is also called the Chebyshev norm
- For a 2D vector $\mathbf{x} = [x_1, x_2]^T$, the L^∞ norm is $\|\mathbf{x}\|_\infty = \max(|x_1|, |x_2|)$
- Example: $\mathbf{a} = [3, -4]^T$ has L^∞ norm $\max(3, 4) = 4$; $\mathbf{b} = [0, 5]^T$ has L^∞ norm $\max(0, 5) = 5$ — so by the L^∞ norm, $\mathbf{b}$ is larger

Figure 3 · Planar Iso-Distance Line of the 2D L_∞ Norm

- The L_∞ norm's iso-distance line is a square, satisfying $\max(|x_1|, |x_2|) = c, c \geq 0$
- Positive infinity is written `numpy.inf` in NumPy — try writing the code that plots this figure yourself



The Lp Norm: a Unified Generalization from L1 to L ∞

$$\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p + \dots + |x_n|^p)^{1/p}, p \geq 1$$

- At $p = 1, 2$, the Lp norm reduces to the L1, L2 norm; as p tends to $+\infty$, the Lp norm tends to the L ∞ norm
- For a 2D vector $\mathbf{x} = [x_1, x_2]^T$, the Lp norm is $(|x_1|^p + |x_2|^p)^{1/p}$
- As p changes, the corresponding iso-distance line on the plane changes shape continuously

Figure 4 · Different Values of p Produce Different Shapes

- As p increases from 1 toward $+\infty$, the contour morphs from a rotated square (rhombus), through a circle, and finally into a square
- The larger p , the sharper and more square-like the corners; the circle at $p = 2$ is the only shape that is fully "isotropic" in every direction

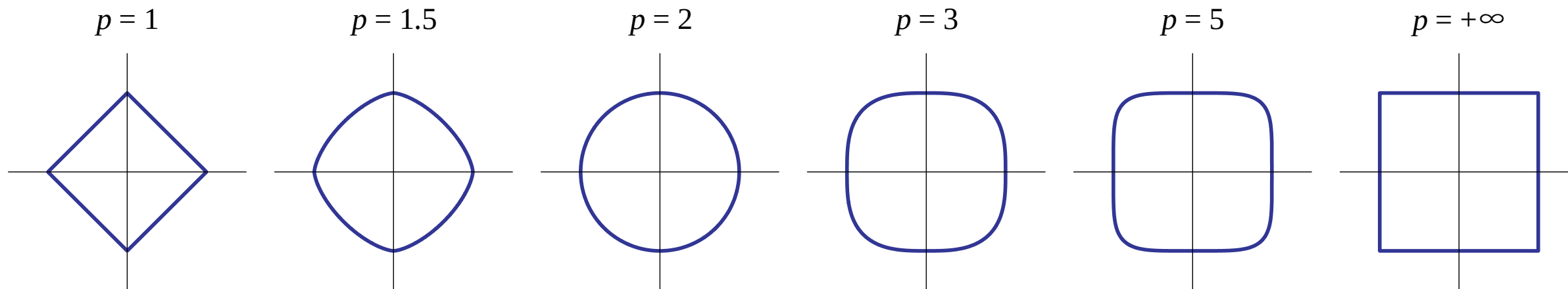
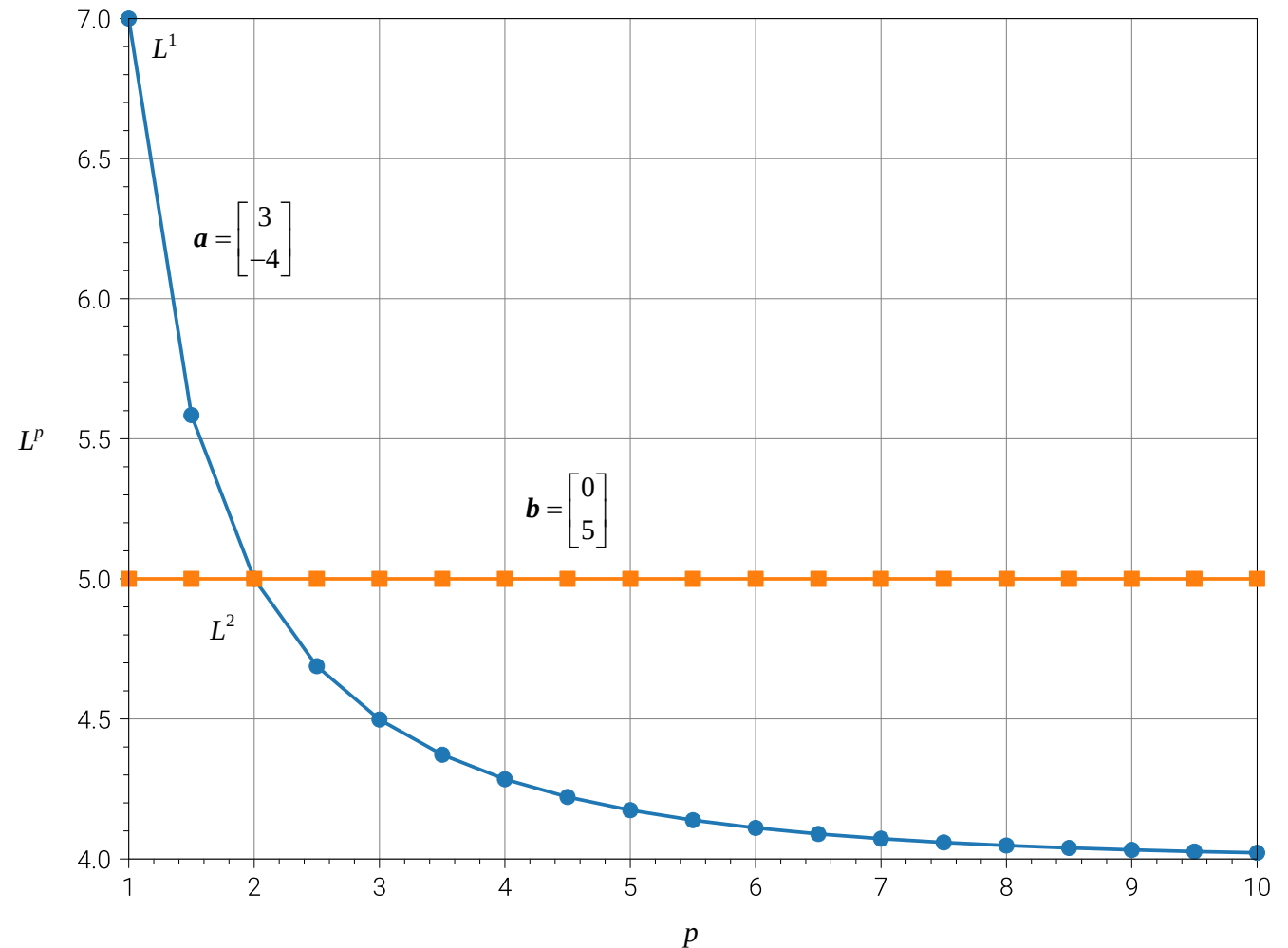


Figure 5 · Lp Norms of Vectors $\mathbf{a}$ and $\mathbf{b}$ as p Varies

- The Lp norm of $\mathbf{a} = [3, -4]^T$ keeps shrinking as p grows: near $p = 1$ every component contributes, while as $p \rightarrow +\infty$ only the largest-magnitude component survives
- $\mathbf{b} = [0, 5]^T$ has only one nonzero component, so its Lp norm stays fixed at 5 regardless of p



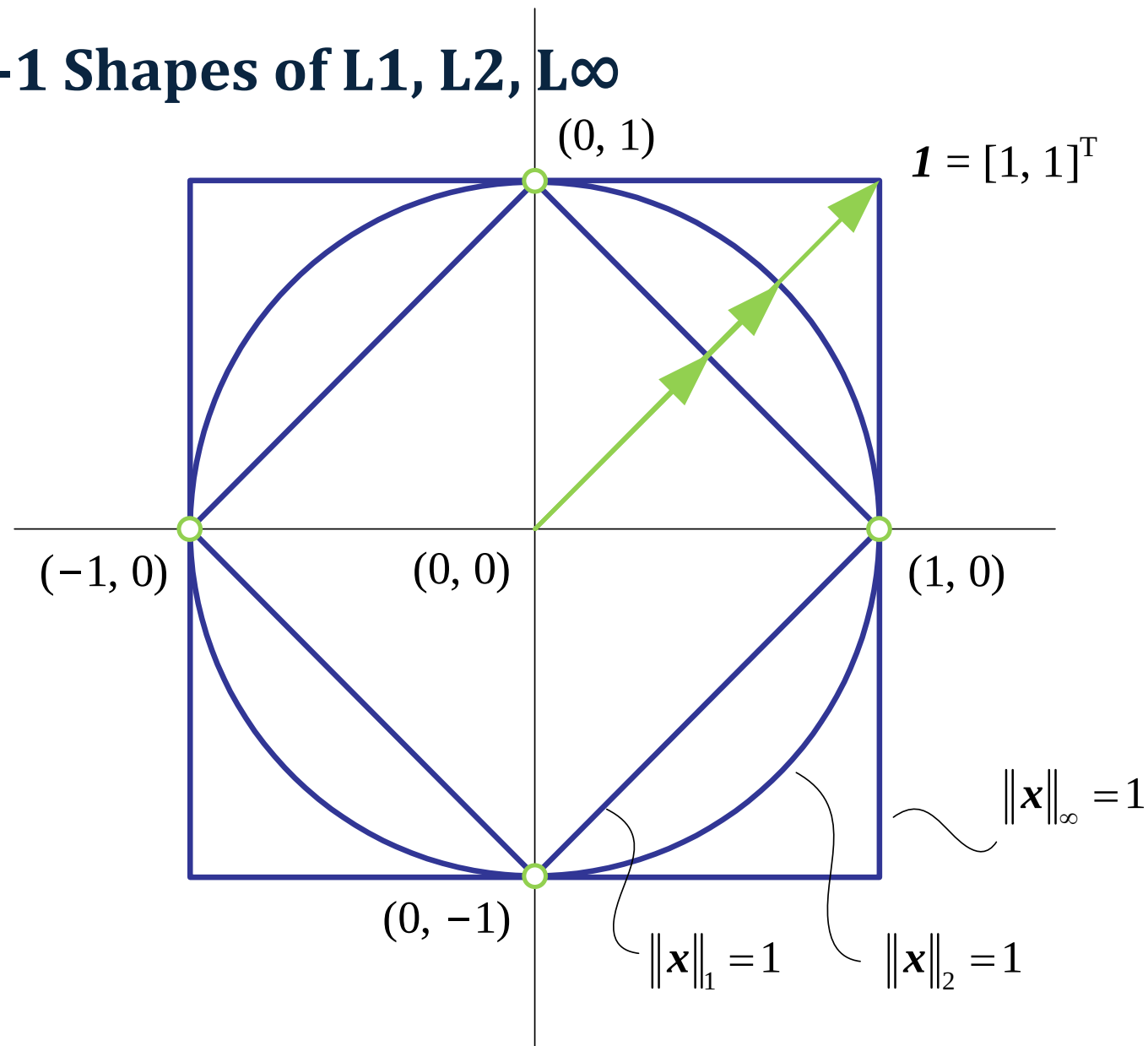
The L1, L2, L ∞ Norm-1 Contours Meet at Four Points

$$\|\mathbf{x}\|_1 = \|\mathbf{x}\|_2 = \|\mathbf{x}\|_\infty = 1 \text{ at } (\pm 1, 0), (0, \pm 1)$$

- The norm-1 contours of L1, L2, L ∞ (rhombus, circle, square) all pass through (1,0), (0,1), (-1,0), (0,-1), where all three norms agree
- Away from these four points, at the same location, the L p norm keeps shrinking as p grows
- Example: the all-ones vector $\mathbf{1} = [1, 1]^T$ has L1 norm 2, L2 norm $\sqrt{2}$, and L ∞ norm 1 — decreasing in that order

Figure 6 · Overlaying the Norm-1 Shapes of L1, L2, L ∞

- The rhombus ($L_1 = 1$), circle ($L_2 = 1$), and square ($L_\infty = 1$) all meet at $(1,0)$, $(0,1)$, $(-1,0)$, $(0,-1)$
- The green arrow points to $[1,1]^T$: away from these four crossing points, the three norms clearly diverge



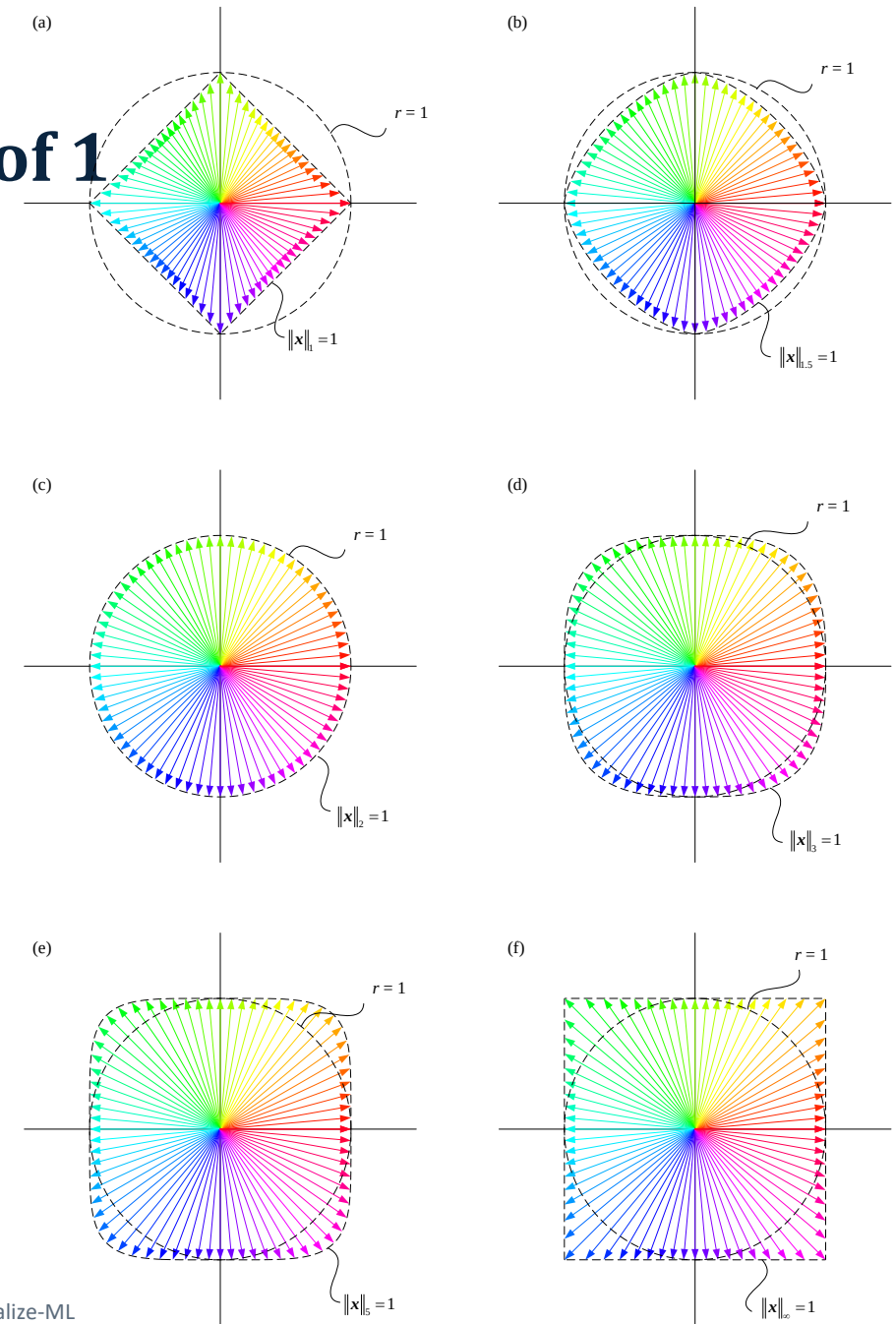
Fix the Lp Norm at 1 and Watch the Endpoints Move

$\|x\|_p = 1$, as p varies continuously from 1 to $+\infty$

- Holding the Lp norm fixed at 1, we watch how the endpoints of the qualifying vectors trace out different curves as p changes
- Only the four axis-aligned directions $(1,0)$, $(0,1)$, $(-1,0)$, $(0,-1)$ keep a length (L2 norm) of exactly 1; every other direction's length shifts with p

Figure 7 · Vectors Satisfying an Lp Norm of 1

- (a) With L1 norm 1, endpoints trace a rotated square (rhombus); lengths (L2 norm) are uneven, and only the four axis points reach length 1
- (b)(c) At L1.5 and L2, the corners soften; at L2 the endpoints fall exactly on the unit circle, so all lengths are equal
- (d)(e)(f) As p grows to 3, 5, $+\infty$, the contour presses ever closer to a square with steeper edges, and lengths elsewhere all exceed 1



A Different Angle: Fix Unit Vectors, Compare Lp Norms

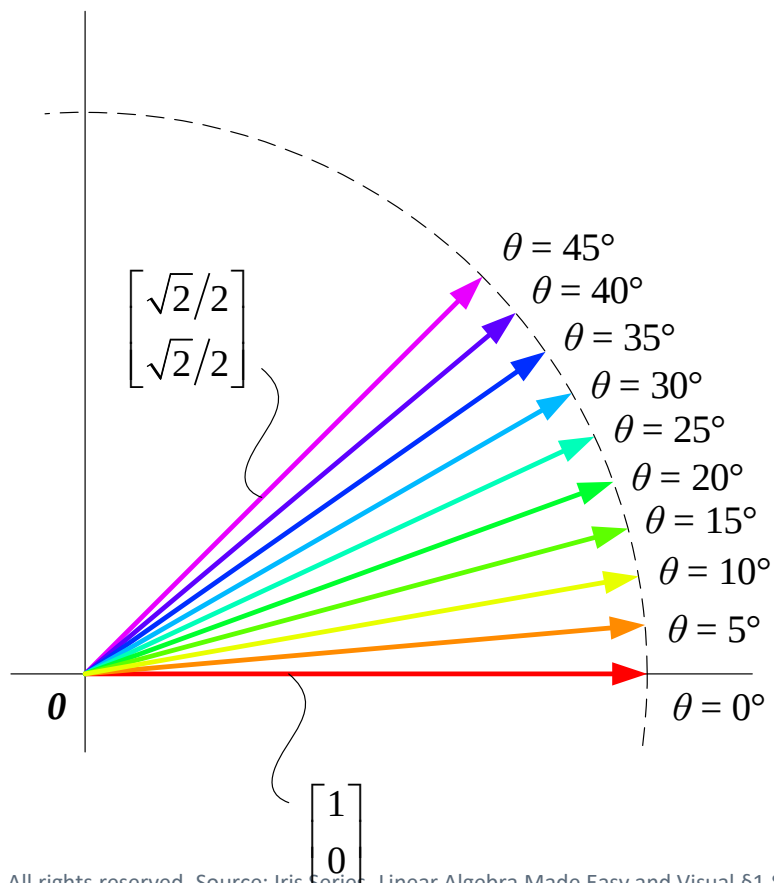
x is a unit vector ($\|x\|_2 = 1$); watch how $\|x\|_p$ varies with p

- This set of unit vectors sits in the first quadrant, with angle to the horizontal sweeping continuously from 0° to 45°
- $[1, 0]^T$ has an Lp norm fixed at 1 regardless of p ; $[\sqrt{2}/2, \sqrt{2}/2]^T$ is the most sensitive to changes in p
- As $p \rightarrow 1$, the Lp norm of $[\sqrt{2}/2, \sqrt{2}/2]^T$ tends to $\sqrt{2}$; as $p \rightarrow +\infty$ it tends to $\sqrt{2}/2$; at $p = 2$ it is exactly 1

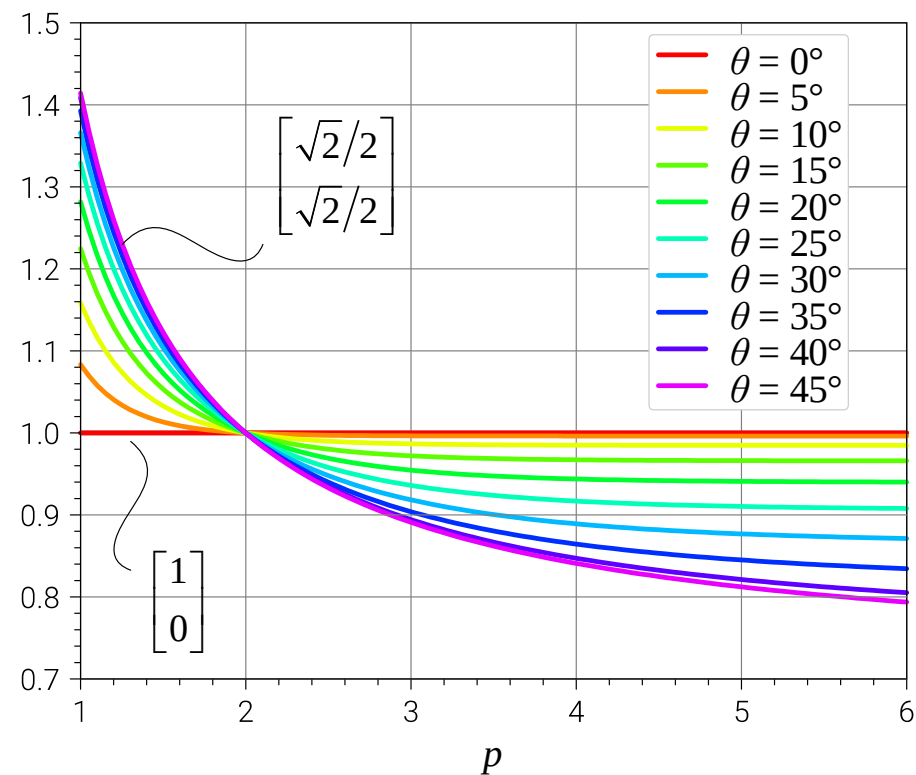
Figure 8 · Lp Norms of Unit Vectors as p Varies

- (a) A set of unit vectors on the plane, angles evenly spaced from 0° to 45° , all with L2 norm 1
- (b) The red curve (0°) keeps an Lp norm of 1; the purple curve (45°) is the "steepest", most sensitive to changes in p

(a)



(b)



$$0 < p < 1$$

Is This Still a Norm When $0 < p < 1$?

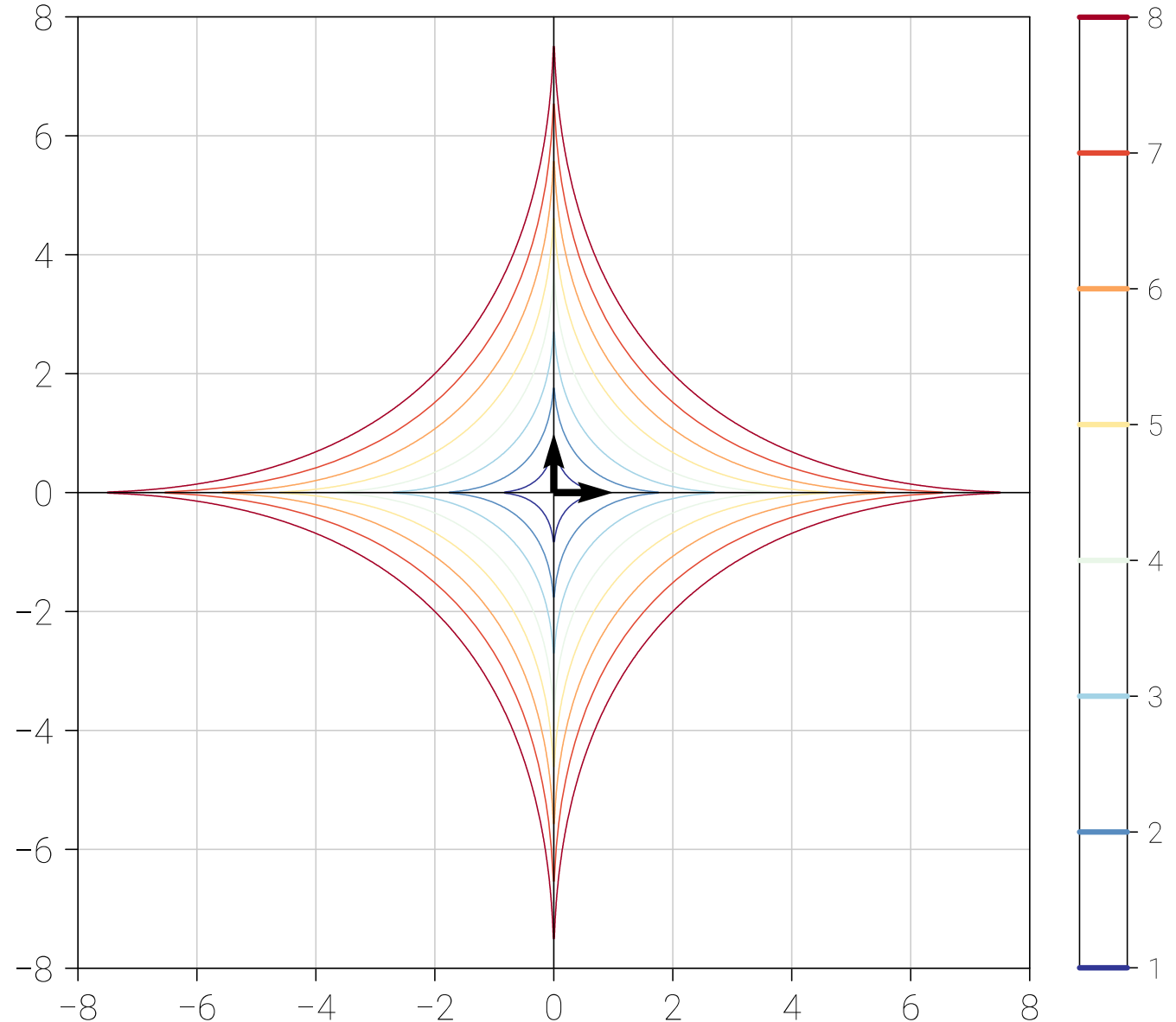
$$\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p)^{1/p}, \quad 0 < p < 1$$

- This expression can still be plotted as a contour when $0 < p < 1$, but it no longer satisfies the definition of a norm (it fails the triangle inequality, i.e. subadditivity)
- The contour becomes "sharp", taking on a star-like shape
- This geometry is not a norm in the strict sense, but it still has value in certain optimization problems and data transformations

$0 < p < 1$

Figure 9 · Contour Line, $p = 0.5$

- At $p = 0.5$, the contour forms a four-petal star, curving noticeably inward toward the axes
- Compared with Figures 1, 2, 3, this makes the essential geometric difference between $p < 1$ and $p \geq 1$ vivid



The Lp Norm of a Three-Dimensional Vector

$$\mathbf{x} = [x_1, x_2, x_3]^T, \|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p + |x_3|^p)^{1/p}$$

- The L1 norm's iso-distance surface is a rotated cube (a special octahedron)
- The L2 norm's iso-distance surface is a sphere; the L^∞ norm's is a cube
- The geometric intuition of the Lp norm carries over naturally from planar contours to spatial surfaces in higher dimensions

Figure 10 · 3D Lp-Norm Iso-Distance Surfaces as p Varies

- At $p = 1, 1.5, 2, 3, 5, 20$, the surface morphs from an octahedron, through a sphere, and approaches a cube
- This mirrors exactly the 2D pattern — only "lines" have become "surfaces"

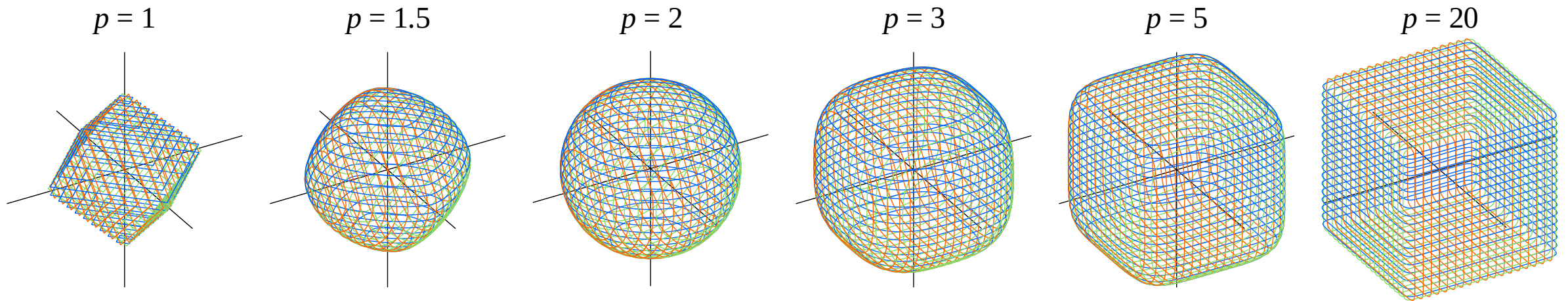
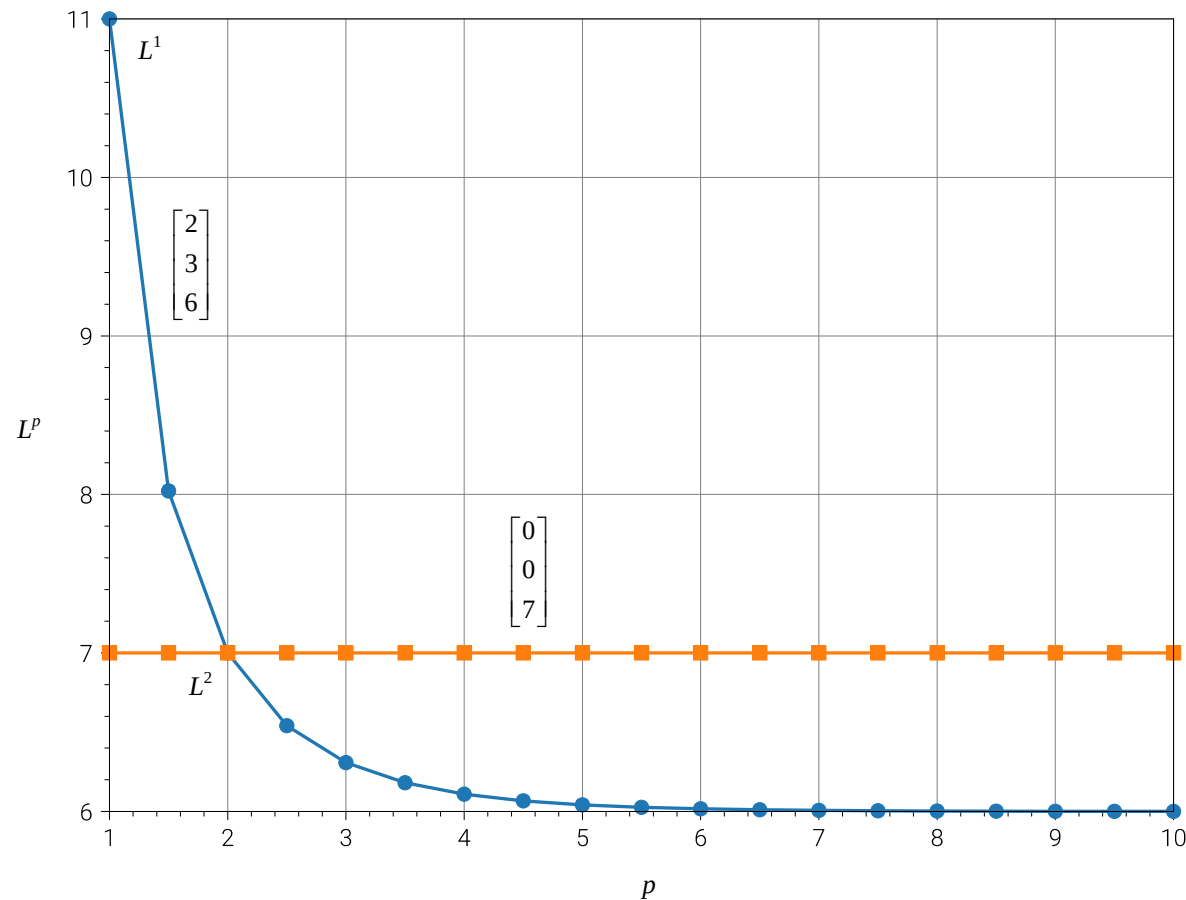


Figure 11 · Lp Norms of $[2,3,6]^T$ and $[0,0,7]^T$ as p Varies

- The Lp norm of $[2, 3, 6]^T$ keeps shrinking as p grows; $[0, 0, 7]^T$ has only one nonzero component, so its Lp norm stays fixed at 7
- Try computing the L1, L2, L^∞ norms of both vectors yourself, and check them against the curve's two end values



Lp Norms of RGB Color Vectors

The red, green, blue vectors e_1, e_2, e_3 keep an L_p norm of 1, unaffected by p

- For yellow, magenta, cyan (e.g. $[1,1,0]^T$), the L_p norm varies with p : it tends to 2 as $p \rightarrow 1$, and to 1 as $p \rightarrow +\infty$
- For white $[1,1,1]^T$, the L_p norm tends to 3 as $p \rightarrow 1$, and to 1 as $p \rightarrow +\infty$

Figure 12 · L^p Norms of Common RGB Color Vectors

- (a) Common color vectors in the RGB cube: red, green, blue, yellow, magenta, cyan, white
- (b)(c)(d) L^p -norm-vs- p curves for the "single-component" colors, "two-component" colors, and white, respectively

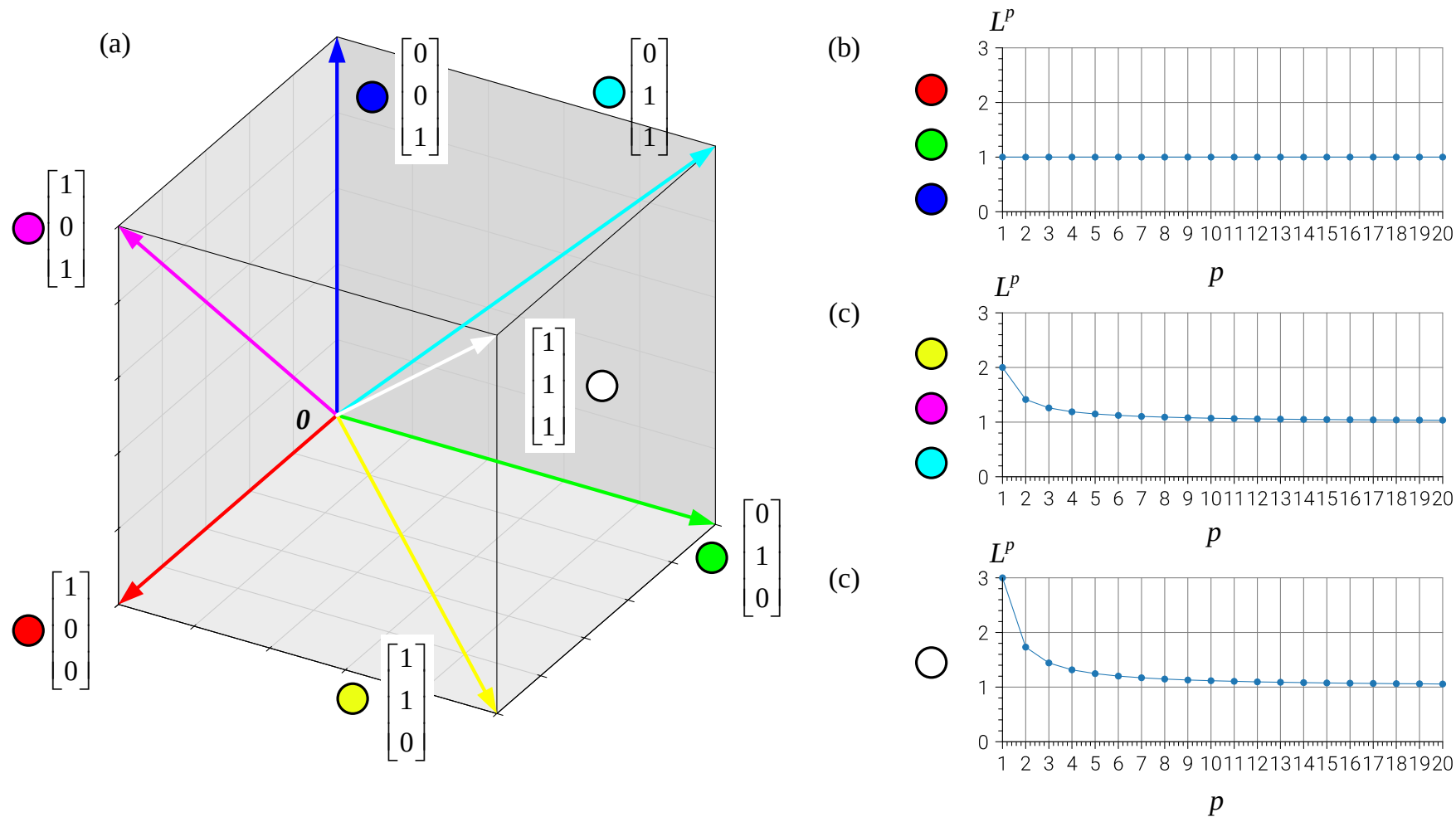


Figure 13 · RGB Color Vectors with L1 Norm 1

- The endpoints of this set of vectors all lie on the same plane, satisfying $x_1 + x_2 + x_3 = 1$
- This is exactly the triangular plane formed where the L1 norm's iso-distance surface intersects the first octant

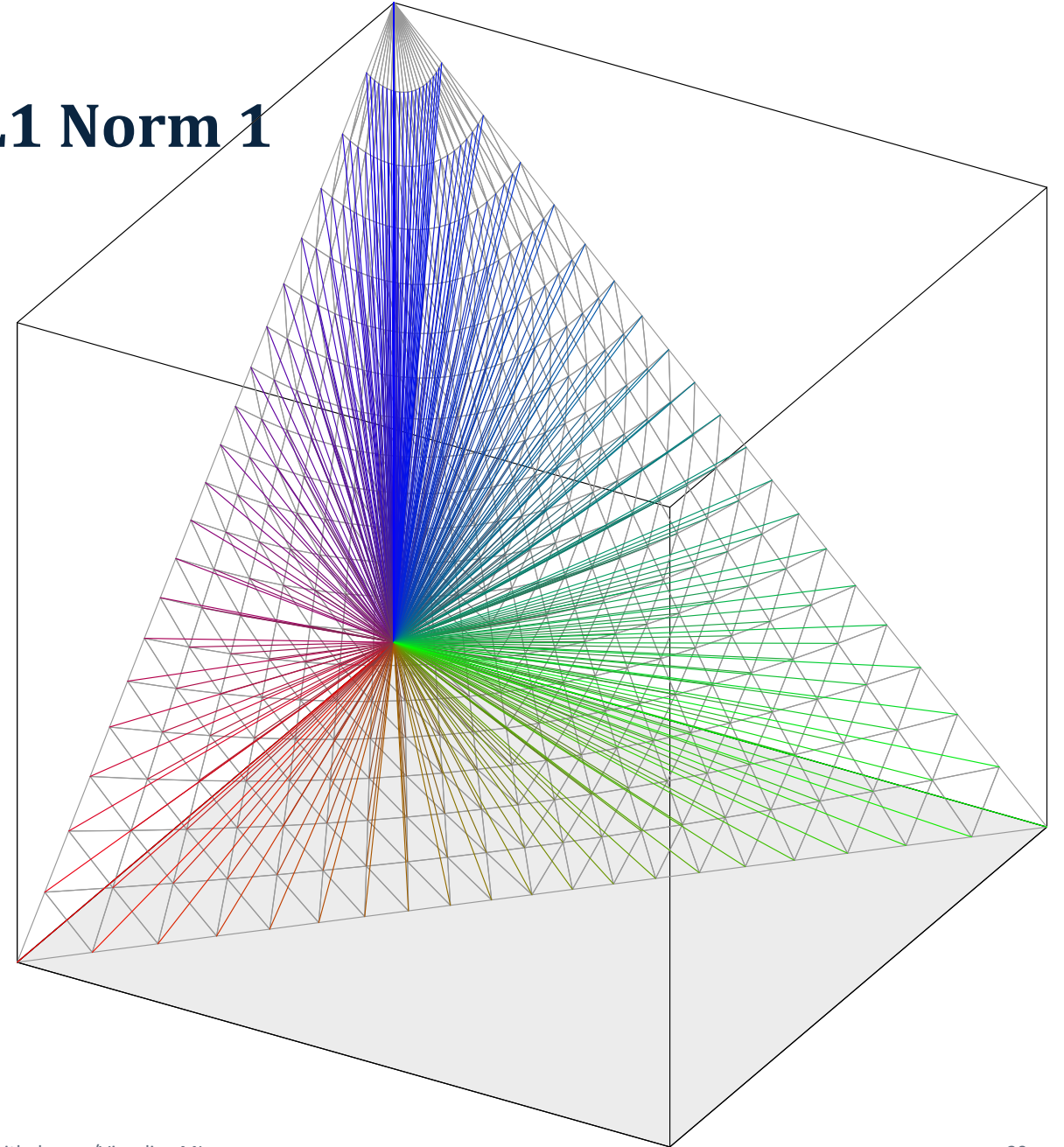


Figure 14 · RGB Color Vectors with L2 Norm 1

- The endpoints of this set of vectors lie on the $1/8$ spherical patch of the unit sphere within the first octant
- This matches our most familiar geometric definition of "length 1" — a unit vector in the Euclidean sense

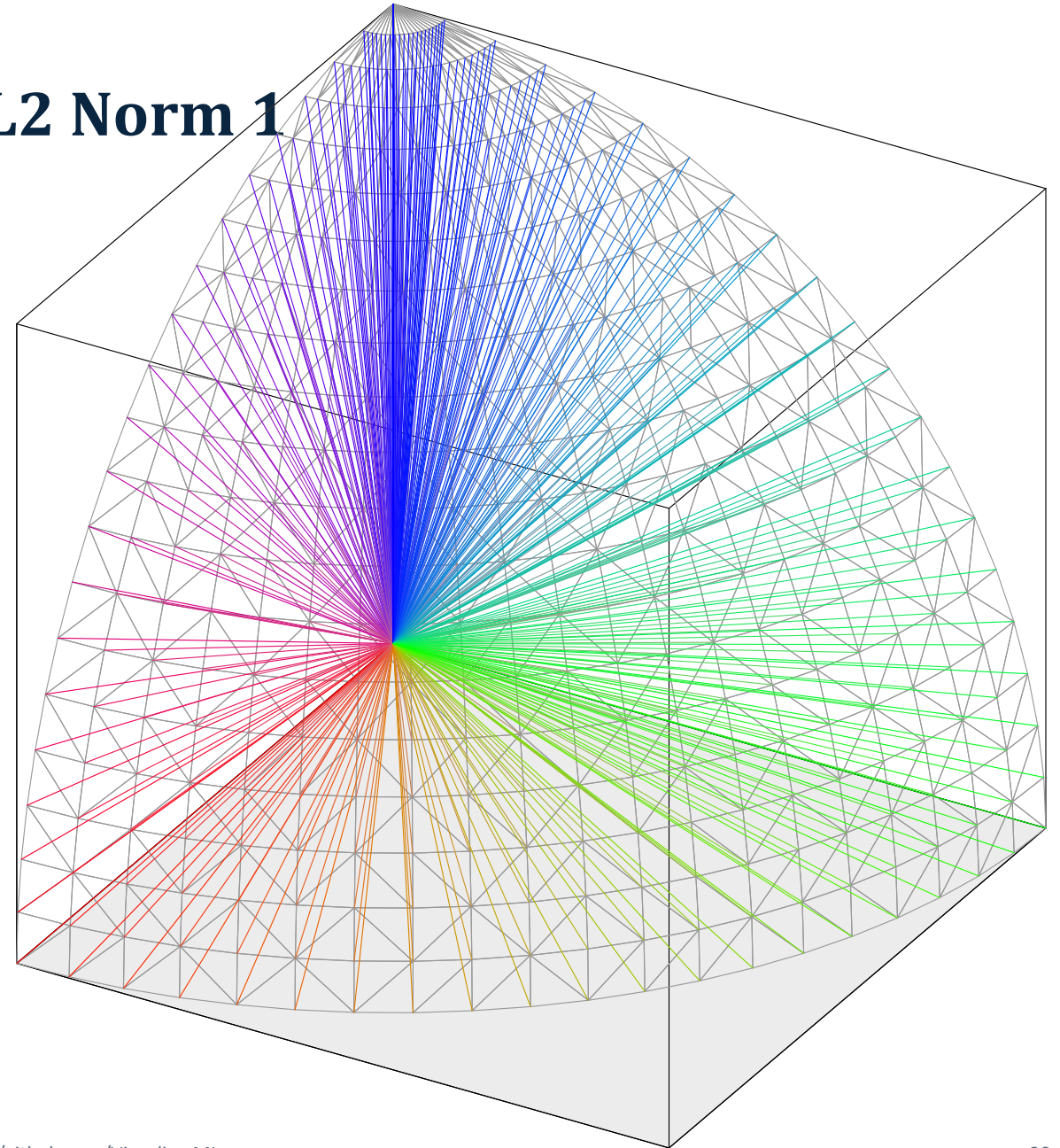
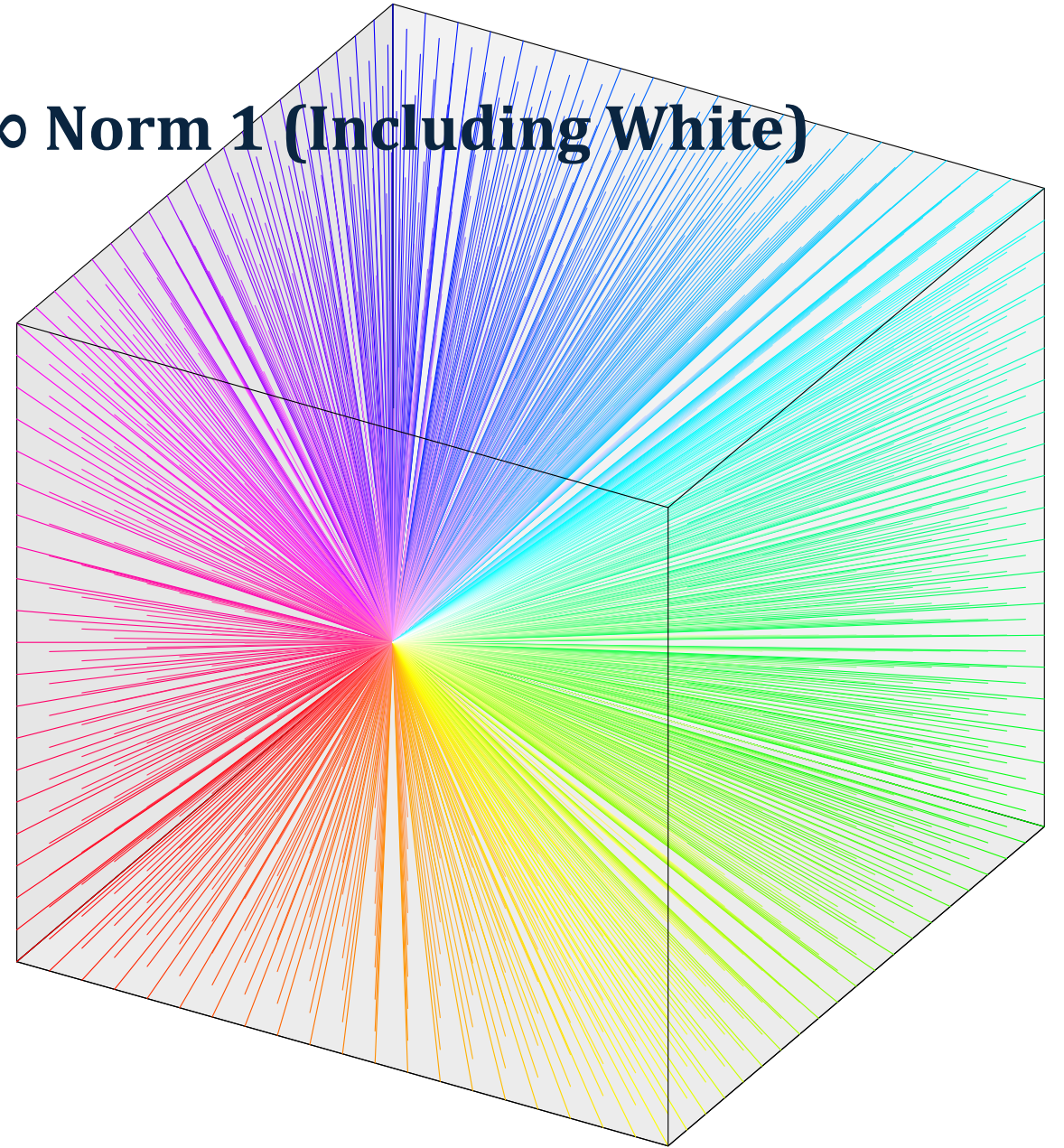


Figure 15 · RGB Color Vectors with L^∞ Norm 1 (Including White)

- The endpoints of this set of vectors all lie on the three brightest faces of the RGB color cube, white included
- The L^∞ norm only cares about the single largest-magnitude component, so every point on those cube faces has norm 1



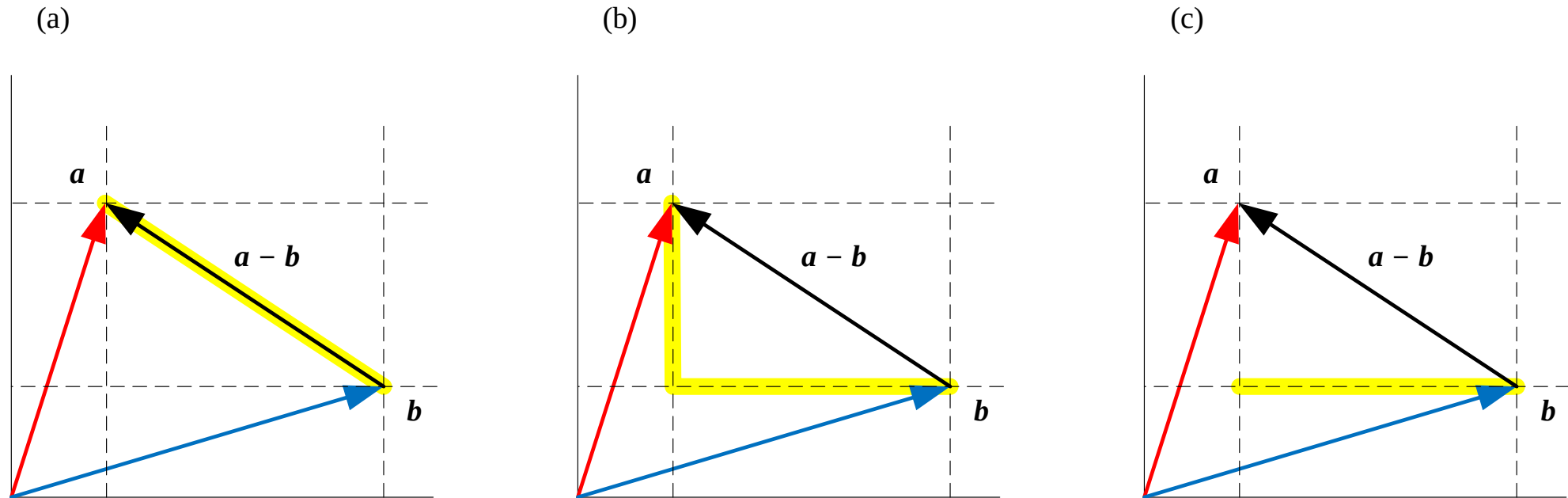
A Vector Norm Defines a Distance Between Vectors

$$\text{distance}(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|$$

- Given two vectors $\mathbf{a}, \mathbf{b}$ on the plane, both starting at the origin unless stated otherwise, "distance" here means the gap between their endpoints
- Choosing a different norm gives a different notion of distance: the L2 norm gives Euclidean distance, the L1 norm gives Manhattan distance, the L^∞ norm gives Chebyshev distance

Figure 16 · Three Different Distance Metrics

- (a) The L2 norm measures the distance between a and b — the everyday, intuitive "straight-line distance"
- (b) The L1 norm measures distance as if walking city blocks: only horizontal and vertical moves allowed
- (c) The L^∞ norm measures distance as whichever single direction has the larger offset



A Matrix That Stores Every Pairwise Distance at Once

26 points, all pairs considered: $C(26, 2) = 325$ distance values

- Place 26 points on the plane, labeled A through Z; here B and S coincide, and so do D and O
- Recording that many distance values one by one isn't practical — a matrix is the natural, compact way to store them
- The next chapter formally introduces the "matrix": it not only stores pairwise distances like these, but also plays a central role in linear transformations

Figure 17 · Pairwise Euclidean Distances Among 26 Points

- Line color depth encodes how far apart two points are; the pairs B/S and D/O coincide exactly
- Image adapted from Coding Is Not Hard

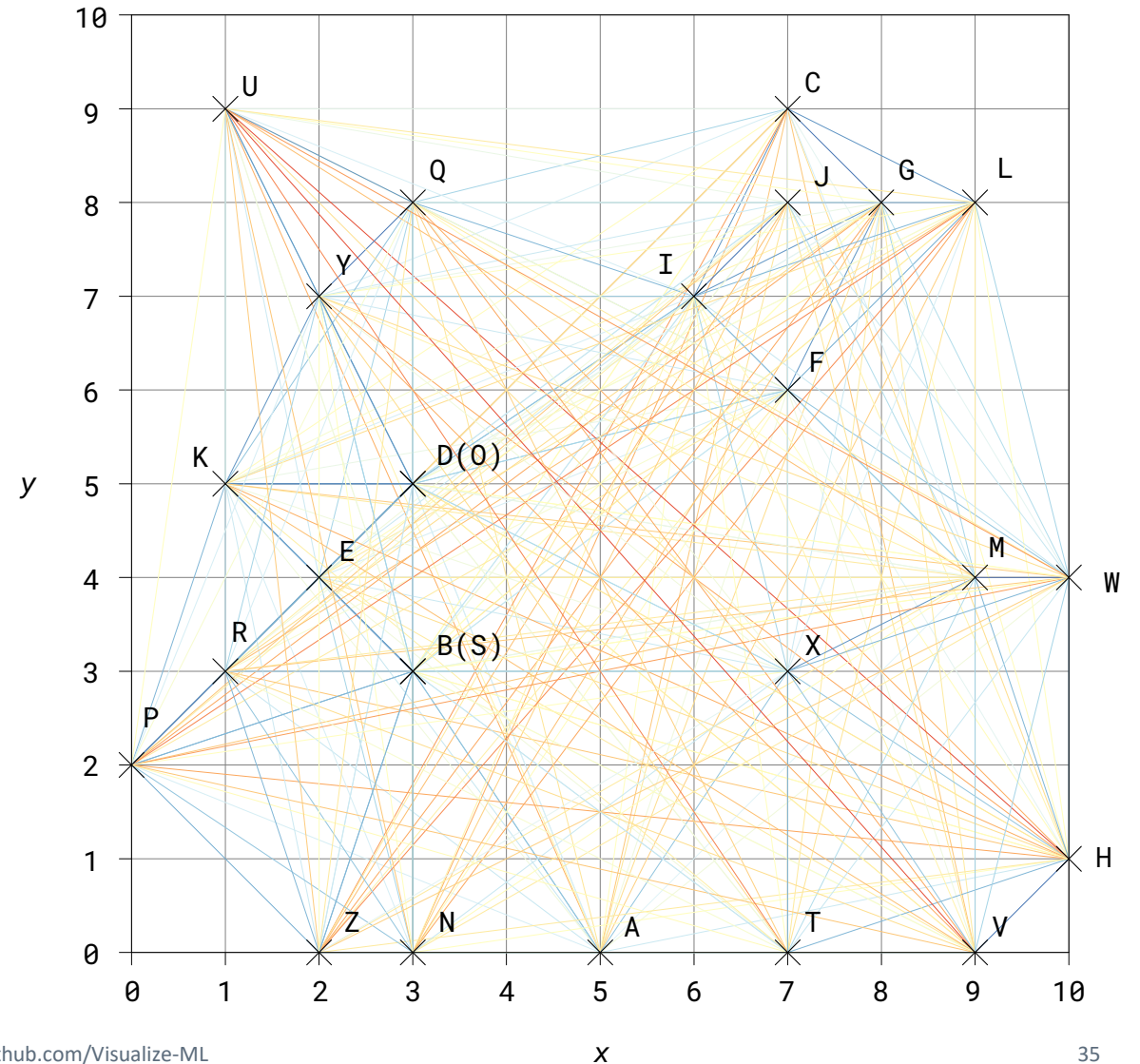
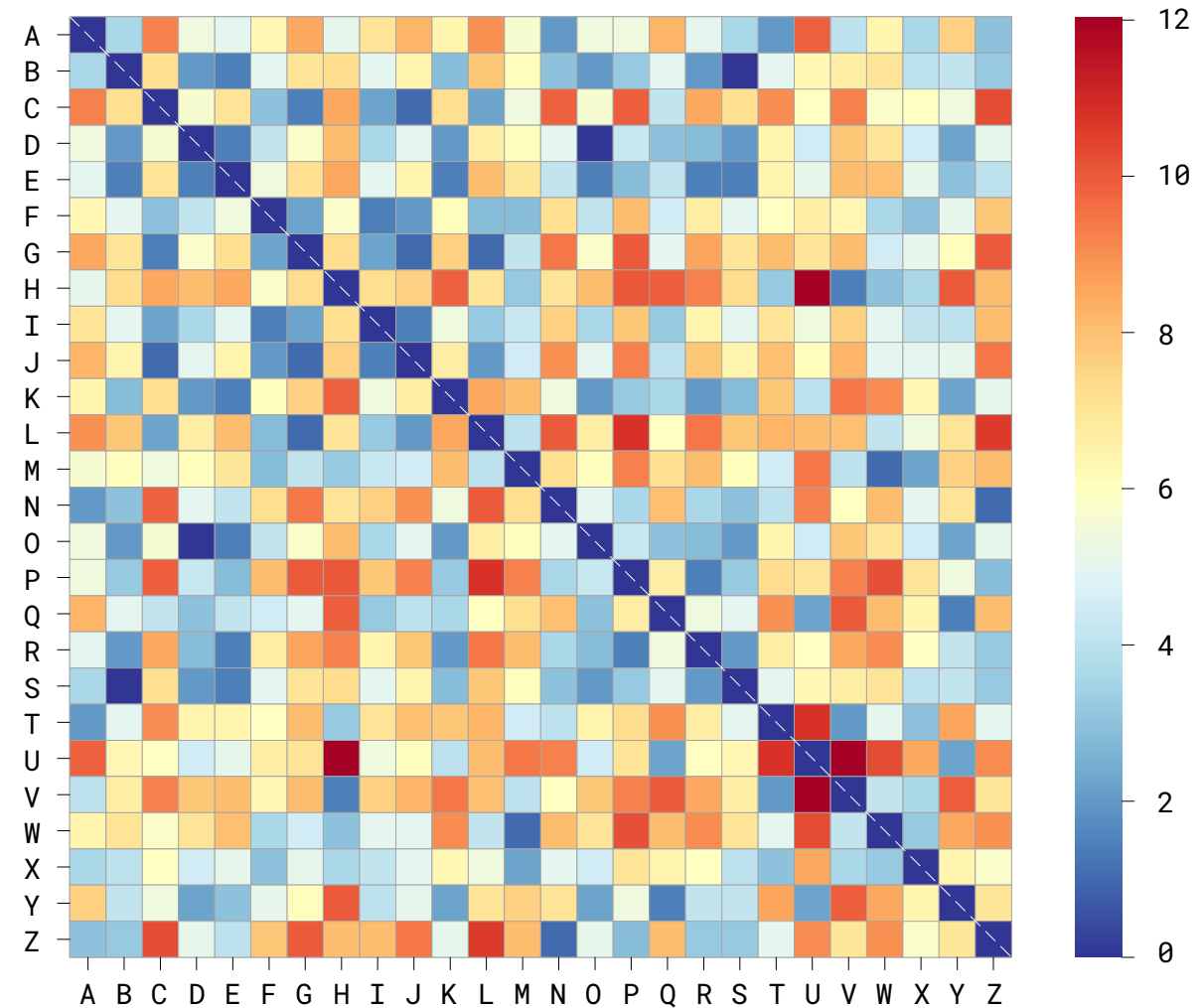


Figure 18 · Pairwise Euclidean Distance Matrix

- The entry in row i , column j gives the Euclidean distance between point i and point j
- Every diagonal entry is 0 (a point's distance to itself), and the matrix is symmetric about the diagonal



Key Terms in This Section

Vector Norm *A unified rule mapping a vector to a comparable scalar*

L2 Norm *Euclidean Norm — Pythagorean theorem, a vector's "length"*

L_p Norm *A continuous, unified generalization from L1 to L_∞, $p \geq 1$*

Subadditivity *The triangle inequality $\|a+b\| \leq \|a\| + \|b\|$ — a required property of any norm*

Manhattan Distance *The "city block" distance defined by the L1 norm*

L1 Norm *Manhattan Distance — sum of the absolute values of the components*

L_∞ Norm *Chebyshev Norm — the largest absolute component*

Iso-Distance Line/Surface *The set of points sharing the same norm value*

Euclidean Distance *The straight-line distance defined by the L2 norm*

Pairwise Distance Matrix *A matrix storing the distance between every pair of points*

FURTHER STUDY

Companion Code & Further Reading

Code 1 · LA_Ch08_01_01.ipynb

Compute a vector's L1 norm, two equivalent ways

Code 2 · LA_01_08_02.ipynb

Plot the L1 norm's planar contour with meshgrid and contour

Further Reading · LA_01_08_03/04.ipynb

Compute and plot the L2, L^∞ norms — worth studying on your own

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Q1 – Q10

Q1 Learn to plot contours with Matplotlib, and explain how they relate to the iso-distance lines in this section.

Q3 Given the vector $[1, 2, 3]^T$, plot how its L_p norm varies with p .

Q5 Write Python code to reproduce Figures 5 and 11.

Q7 Why isn't the expression for $0 < p < 1$ a norm?

Q9 Try using `matplotlib.pyplot.contourf()` to plot filled contours for Figures 1, 2, and 3.

Q2 Given the vector $[1, 2, 3]^T$, compute its L_1 , L_2 , L_∞ norms by hand.

Q4 Compare the L_p norms of each pair below as p varies, and explain why: $[1, 2, 2]^T$ vs $[0, 0, 3]^T$; $[2, 4, 4]^T$ vs $[0, 0, 6]^T$; $[2, 3, 6]^T$ vs $[0, 0, 7]^T$; $[1, 4, 8]^T$ vs $[0, 0, 9]^T$.

Q6 What is subadditivity, and how does it relate to the definition of a vector norm?

Q8 Compute the L_1 , L_2 , L_∞ norms of these color vectors: red $[1, 0, 0]^T$, yellow $[1, 1, 0]^T$, white $[1, 1, 1]^T$.

Q10 Write a function `lp_norm(x, p)` that computes the L_p norm by hand (without calling `np.linalg.norm`), and check it against NumPy's result.

Q11 – Q20

Q11 Using `np.linalg.norm()`, compute the L_1 , L_2 , L_∞ norms of $[1, -2, 3, -4]^T$, and verify that $L_\infty \leq L_2 \leq L_1$.

Q13 In Python, plot the p -norm-1 contour on the plane for $p = 0.5, 1, 1.5, 2, 4, +\infty$ (mirroring Figure 4).

Q15 Use `scipy.spatial.distance`'s `euclidean`, `cityblock`, and `chebyshev` functions, and confirm they match your hand-written L_2 , L_1 , L_∞ distance formulas.

Q17 Use Matplotlib to plot 26 random points with their pairwise Euclidean distance lines (mirroring Figure 17), using color depth to show distance.

Q19 Write a function `rgb_norm_curve(color)` that plots how a given RGB color vector's L_p norm varies with p (mirroring Figures 5 and 12).

Q12 Write code that verifies $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ for randomly generated vectors.

Q14 Write a function `iso_surface_3d(p)` that uses Matplotlib 3D plotting to draw the L_p -norm-1 surface in 3D (mirroring Figure 10).

Q16 Write a function `pairwise_distance_matrix(points, ord)` that takes a set of 2D points and a norm type and returns the pairwise distance matrix (mirroring Figure 18).

Q18 Visualize a pairwise distance matrix with `seaborn.heatmap()` or `matplotlib.pyplot.imshow()`, including a color bar.

Q20 Use Python to verify that the red, green, and blue basis color vectors keep an L_p norm of 1 regardless of p .

Q21 – Q29

Q21 Write a function `is_norm_p(p)` that samples many random vector pairs to check whether the triangle inequality (subadditivity) holds for $0 < p < 1$.

Q23 Use `sklearn.preprocessing.normalize()` to normalize a set of random vectors by the L1 and L2 norm, and check whether the normalized directions change.

Q25 Write a function `unit_vectors_by_angle(n)` generating n unit vectors with angles evenly spaced across 0° – 90° , and plot how their L_p norm varies with p (mirroring Figure 8).

Q27 Generate a random 3D vector, compute its L1, L2, L_∞ norms, and use a 3D scatter plot to mark its position relative to the surfaces in Figure 10.

Q29 Combine what you've learned: write a small program that takes a set of 2D or 3D points and outputs their pairwise L1, L2, L_∞ distance matrices, visualizing one of them as a heatmap.

Q22 Plot the contours for $p = 0.3, 0.5, 0.8$ with Matplotlib, and compare how much sharper the star shape gets as p shrinks.

Q24 The L1 and L2 norms are widely used for regularization in machine learning. Look this up, and explain in a sentence or two why L1 regularization tends to produce sparse solutions.

Q26 Write a function `chebyshev_vs_manhattan(a, b)` comparing the ratio of Chebyshev distance to Manhattan distance for a pair of vectors, and study the distribution of that ratio over many random pairs.

Q28 Write a function `lp_norm_limit(x)` that uses $p = 100, 1000$ to approximately confirm that the L_p norm converges to the L_∞ norm as $p \rightarrow +\infty$.

10 Prompts Worth Exploring (1/2)

- 1 Explain why a vector norm must satisfy subadditivity (the triangle inequality), and what goes wrong if it doesn't.
- 2 Use an everyday example to illustrate the intuitive difference between L_1 , L_2 , and L_∞ norms — say, a taxi's driving distance versus a straight-line distance.
- 3 Prove that as $p \rightarrow +\infty$, the L_p norm's limit really does converge to the L_∞ norm (the largest absolute component).
- 4 Why doesn't the expression for $0 < p < 1$ satisfy the definition of a norm? Give a counterexample based on the triangle inequality.
- 5 Explain where Euclidean, Manhattan, and Chebyshev distance are each best suited — in machine learning, in path planning, and beyond.

10 Prompts Worth Exploring (2/2)

6

Design a small Python experiment showing that the same pair of vectors can have wildly different "distances" under different norms.

7

Explain why L1 regularization in machine learning tends to produce sparse solutions, while L2 regularization tends to shrink weights more evenly.

8

What are common applications of a pairwise distance matrix in data science — clustering, recommendation systems, nearest-neighbor search?

9

Explain why the red, green, and blue basis vectors in RGB space keep an L_p norm of 1, unaffected by p .

10

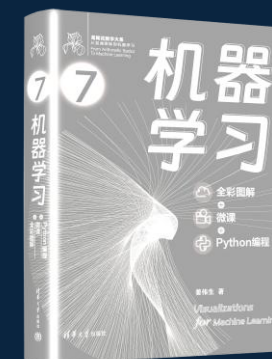
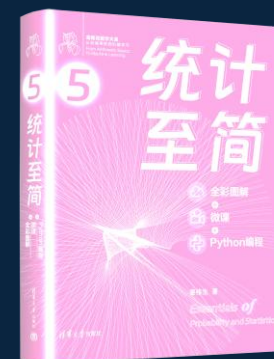
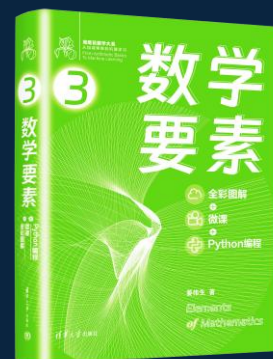
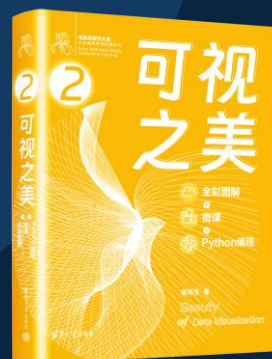
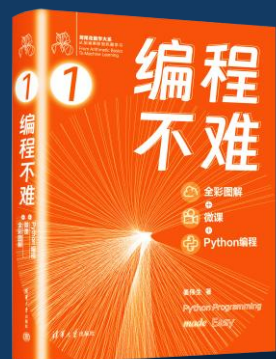
Beyond a 3D scatter plot, what other ways could you visualize the L_p norm's iso-distance surface in three or more dimensions?

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

